

12) Quantiles and Box plots

* Quantiles are values that divide a dataset or probability distribution into equal parts.

25% quantile (First quantile - Q_1)

50% quantile (Median - Q_2)

75% quantile (Third quantile - Q_3)

$$\text{Interquartile range (IQR)} = Q_3 - Q_1$$

Ex: $Q_1 = 18.35$

$Q_2 = 27.8$

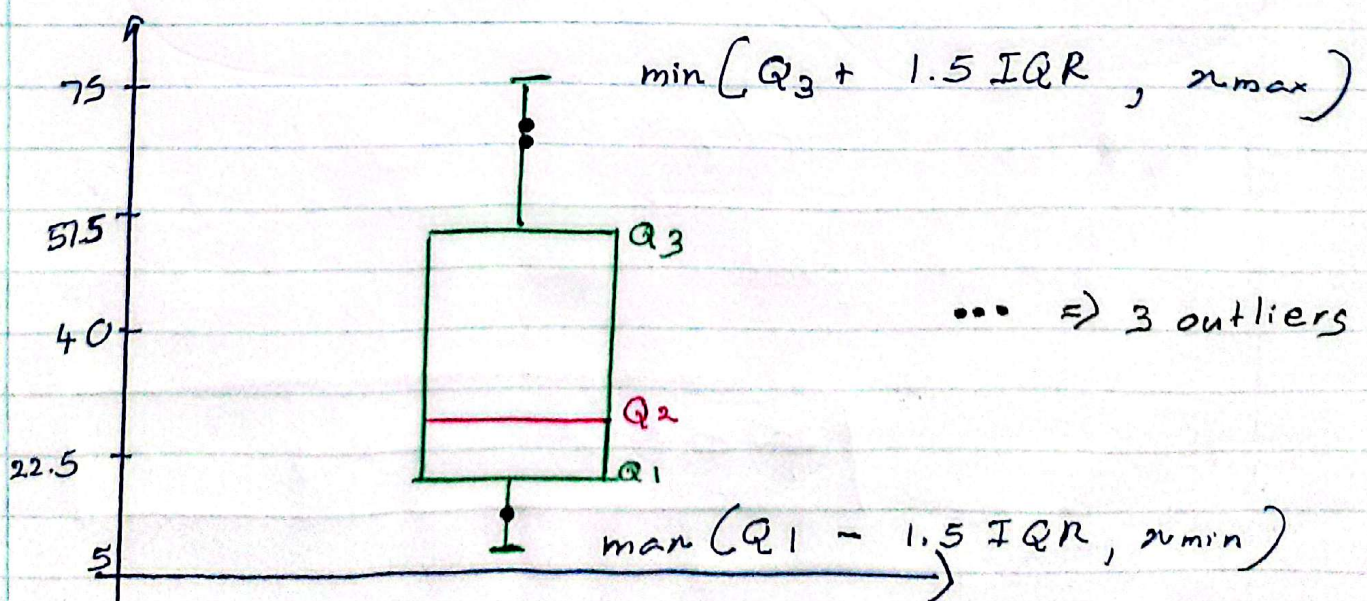
$Q_3 = 52.95$

$IQR = 34.6$

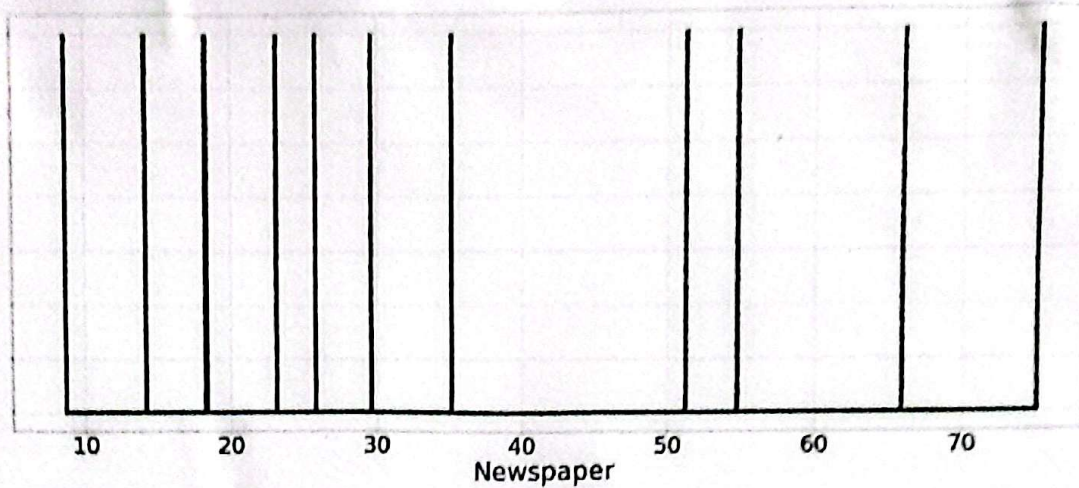
$x_{\min} = 8.7$

$x_{\max} = 75$

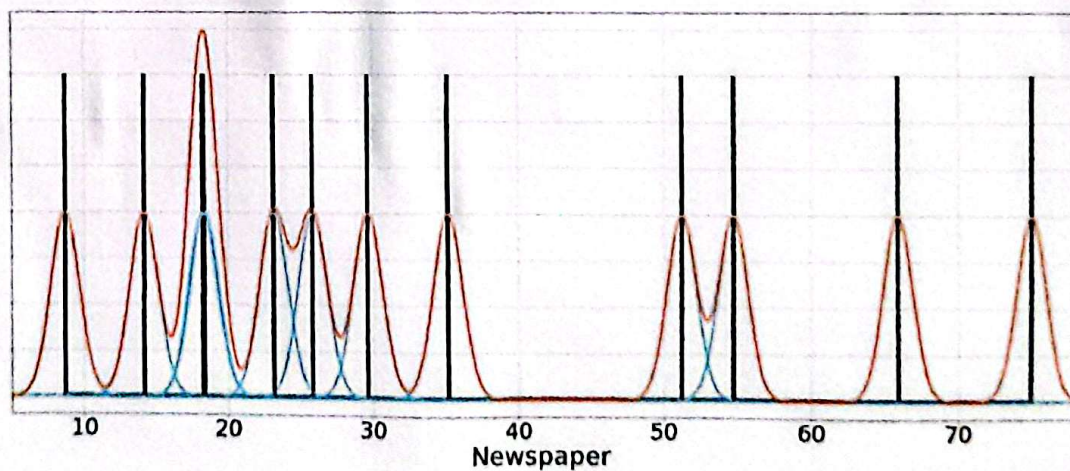
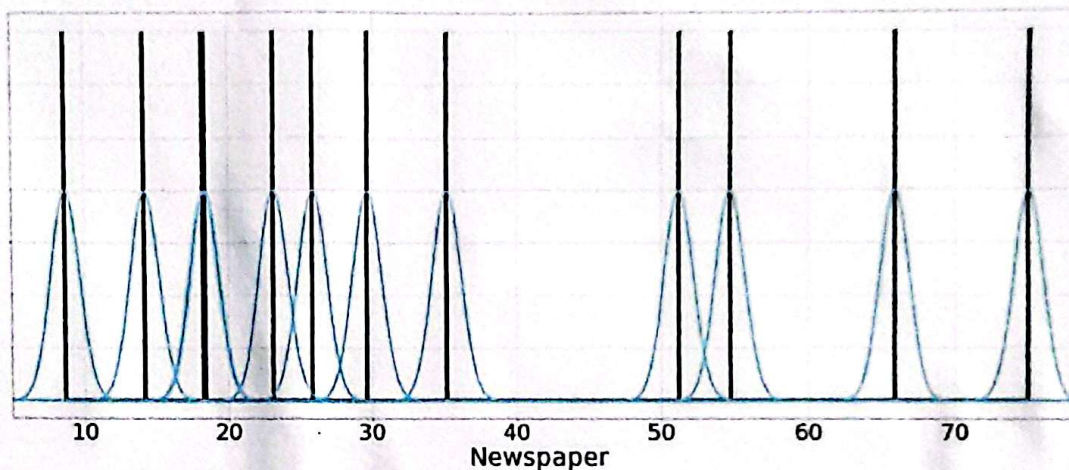
⇓
Box plot



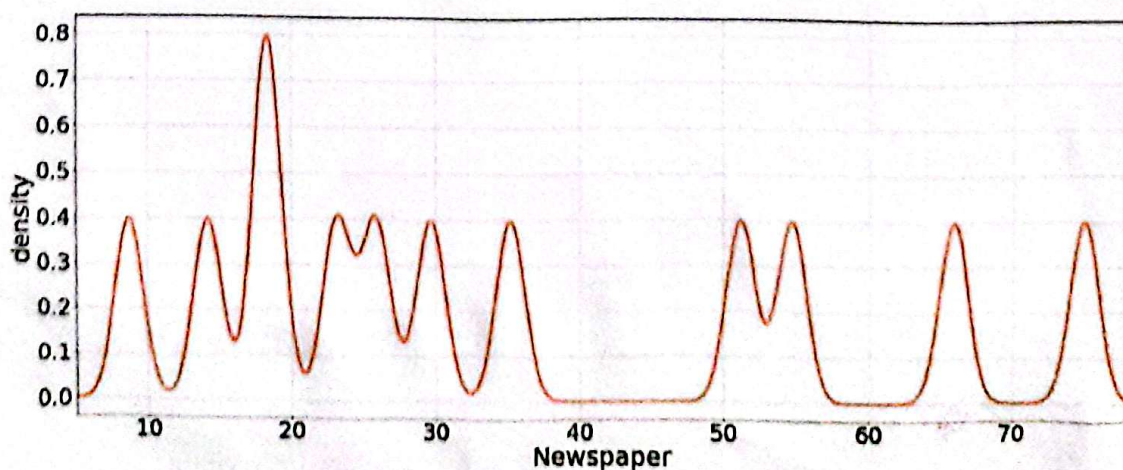
First: draw your observations along the x axis



Second: draw a gaussian centered at each observation

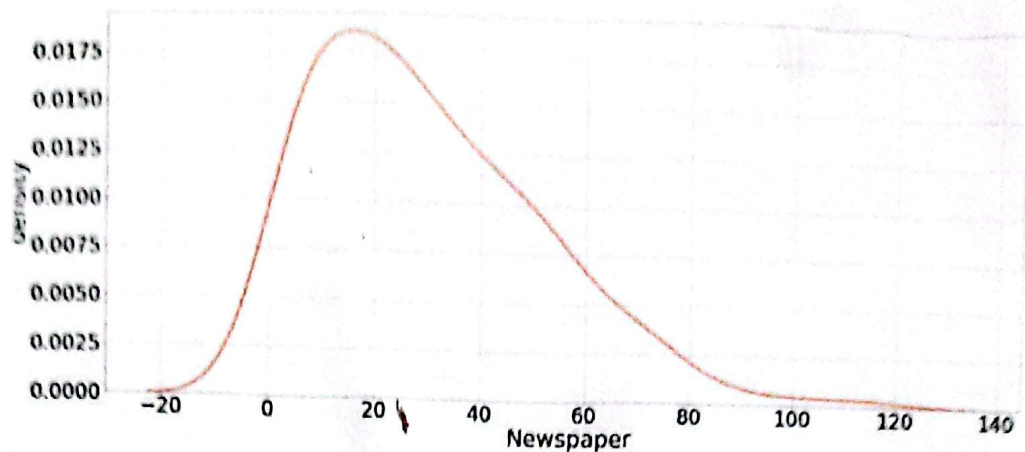


Third: multiply everything by $1/n$ and sum the curves

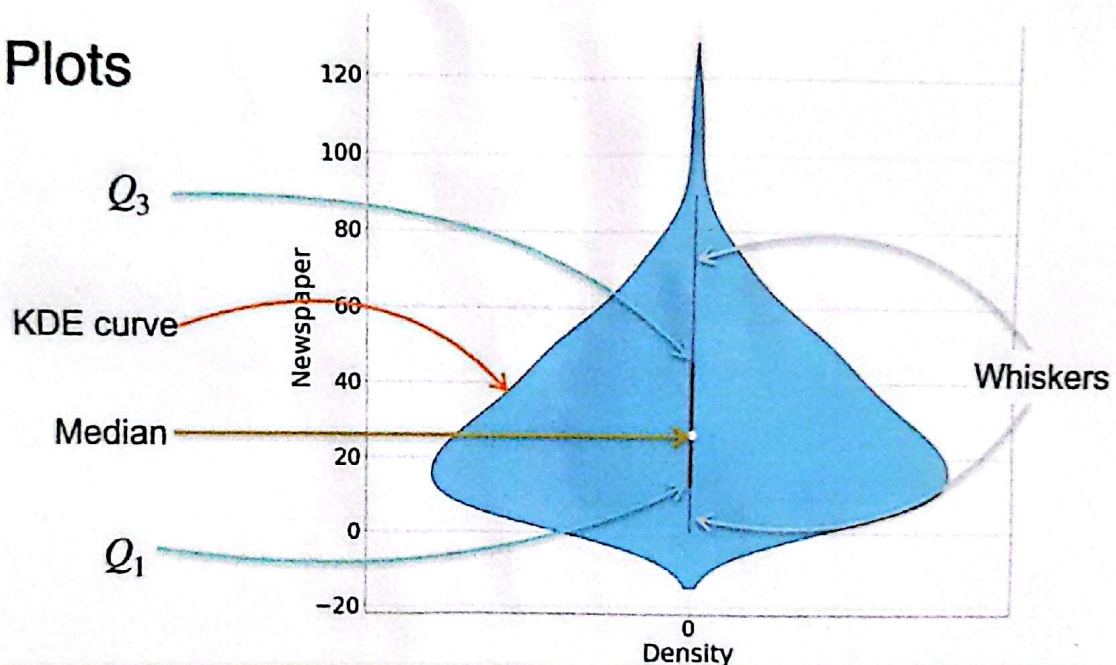


Kernel Density Estimation

What if you used all the dataset?



Violin Plots



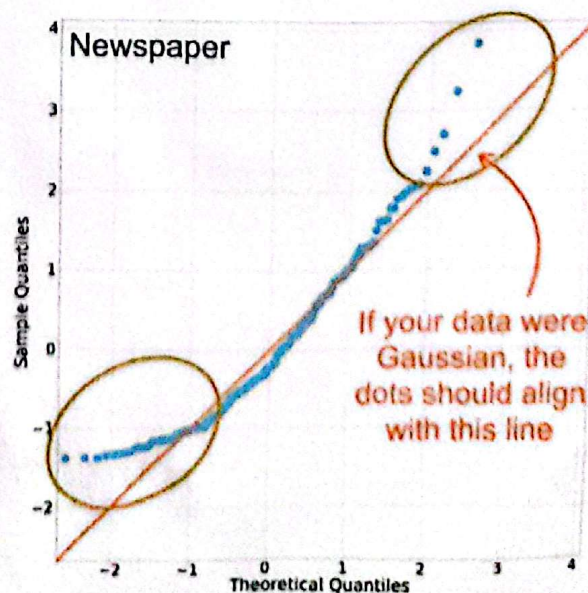
QQ Plots

Quantile-Quantile plots (QQ Plots) compare quantiles

- Standardize your data:

$$\left(\frac{x - \mu}{\sigma} \right)$$

- Compute quantiles
- Compare to gaussian quantiles



13) Discrete Joint distribution

*] A joint distribution tells us how the probability of two (or more) random variables taking on certain values together.

*] $\left. \begin{array}{l} X = \text{random variable 1} \\ Y = \text{random variable 2} \end{array} \right\} \boxed{P(X=x, Y=y)}$

*] For independent discrete random variables

$$\boxed{P_{XY}(x, y) = P(X=x, Y=y) = P(x) \cdot P(y)}$$

*] For discrete we usually represent the joint distribution in a table.

X/Y	y_1	y_2	y_3
x_1	0.10	0.15	0.05
x_2	0.20	0.25	0.05
x_3	0.05	0.10	0.05

*] Each cell = $P(X=x_i, Y=y_j)$

*] All probabilities must be non-negative

*] The sum of probabilities = 1

$$\boxed{\sum_i \sum_j P(X=x_i, Y=y_j) = 1}$$

14) Continuous Joint distributions.

- * Discrete joint distributions assign probabilities to individual outcomes, while continuous joint distributions assign probabilities to ranges or intervals of values

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

example: X between 1 and 2
 Y between 0 and 3

$$P(1 \leq X \leq 2, 0 \leq Y \leq 3) = \int_0^3 \int_1^2 f_{X,Y}(x, y) dy dx$$

15) Marginal Distribution

- * The marginal distribution of a random variable is just its own probability distribution, ignoring other variables
- * If you have a joint distribution of two variables X and Y , the marginal distribution of X tells you the probabilities (densities) of X alone regardless of what value Y takes.
- * It is called marginal because when probabilities are written in a table, you get them by summing along the margins (row/columns)

Discrete

$$P(X=x_i, Y=y_j)$$

$$\Rightarrow \text{Marginal of } X \Rightarrow P(X=x_i) = \sum_j P(X=x_i, Y=y_j)$$

$$\Rightarrow \text{Marginal of } Y \Rightarrow P(Y=y_j) = \sum_i P(X=x_i, Y=y_j)$$

ex :

X/Y	y_1	y_2	Total (marginal of X)
x_1	0.1	0.2	0.3
x_2	0.3	0.4	0.7
Total			<u>1</u>
(marginal of Y)	0.4	0.6	

$$P(X=x_1) \Rightarrow 0.1 + 0.2 = 0.3 //$$

$$P(Y=y_2) \Rightarrow 0.2 + 0.4 = 0.6 //$$

Continuous

$f_{X,Y}(x,y)$:

$$\text{Marginal PDF of } X : f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$\text{Marginal PDF of } Y : f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

16) Conditional Distribution

* The conditional distribution of a random variable tells you how it is distributed when you already know the value of other variable.

Discrete

$$P(X=x_i | Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)}$$

(if $P(Y=y_j) > 0$)

ex :	X/Y	y_1	y_2
	x_1	0.1	0.2
	x_2	0.3	0.4

* Marginal of $Y=y_2 = 0.2 + 0.4 = 0.6$

* Conditional distribution of X given $Y=y_2$;

$$P(X=x_1 | Y=y_2) = \frac{0.2}{0.6}$$

$$P(X=x_2 | Y=y_2) = \frac{0.4}{0.6}$$

Conditional

$$f_{Y|X=x}(y) = \frac{f_{XY}(x,y)}{f_X(x)}$$

Joint PDF of X and Y

marginal of X

17) Covariance of a dataset

$$\text{Cov}(X, Y) = \sum xy \quad \text{Almost ...}$$

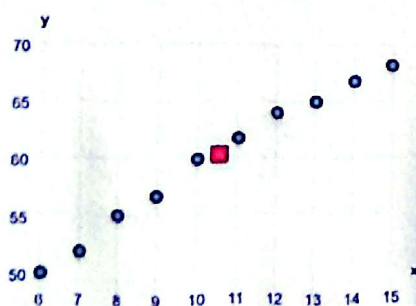
$$\text{Cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

For unequal probabilities

$$\text{Cov}(x, y) = \sum p_{xy}(x_i, y_i)(x_i - \mu_x)(y_i - \mu_y)$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

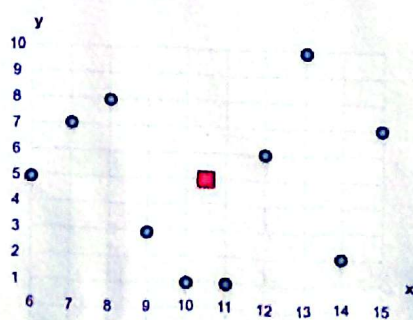
Comparing Correlations



Age vs Height

Covariance > 0

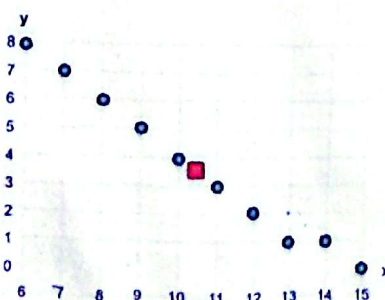
$$\text{Cov}(x, y) = 17$$



Age vs Grades

Covariance ≈ 0

$$\text{Cov}(x, y) = 0.1$$

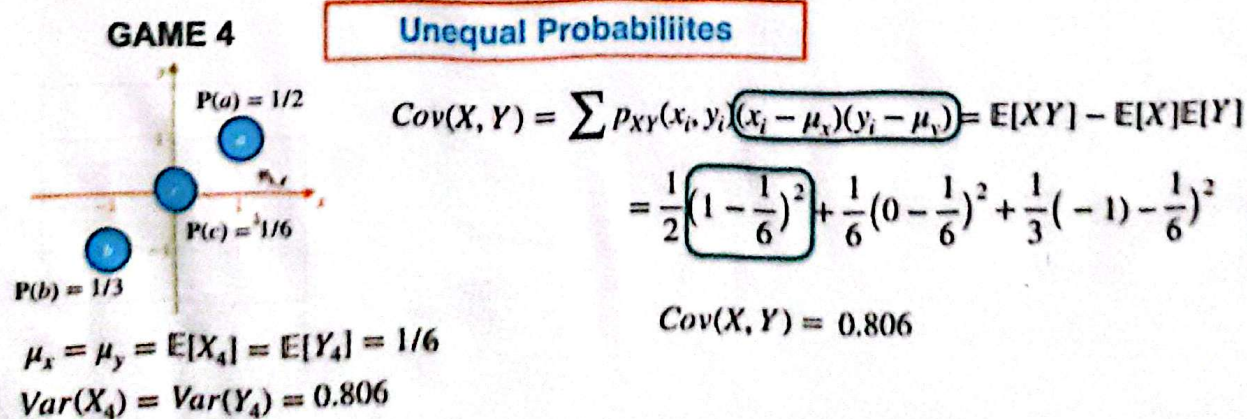
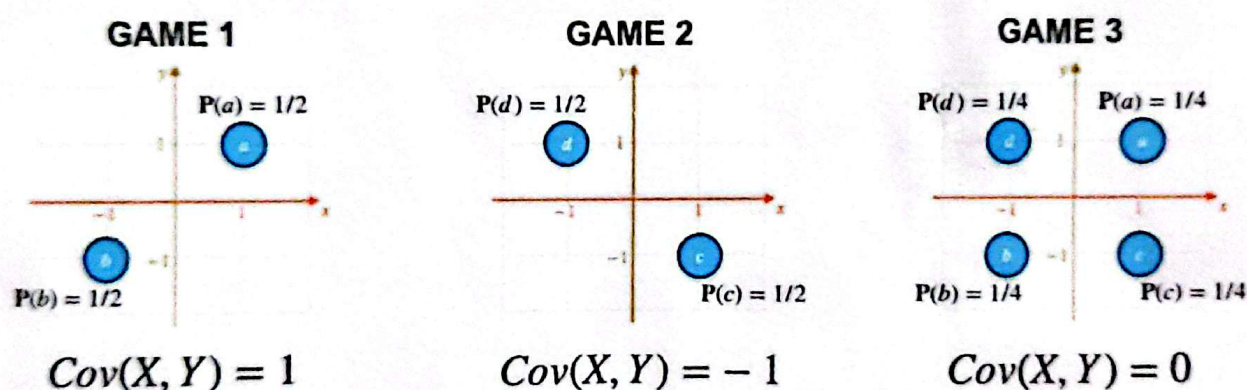
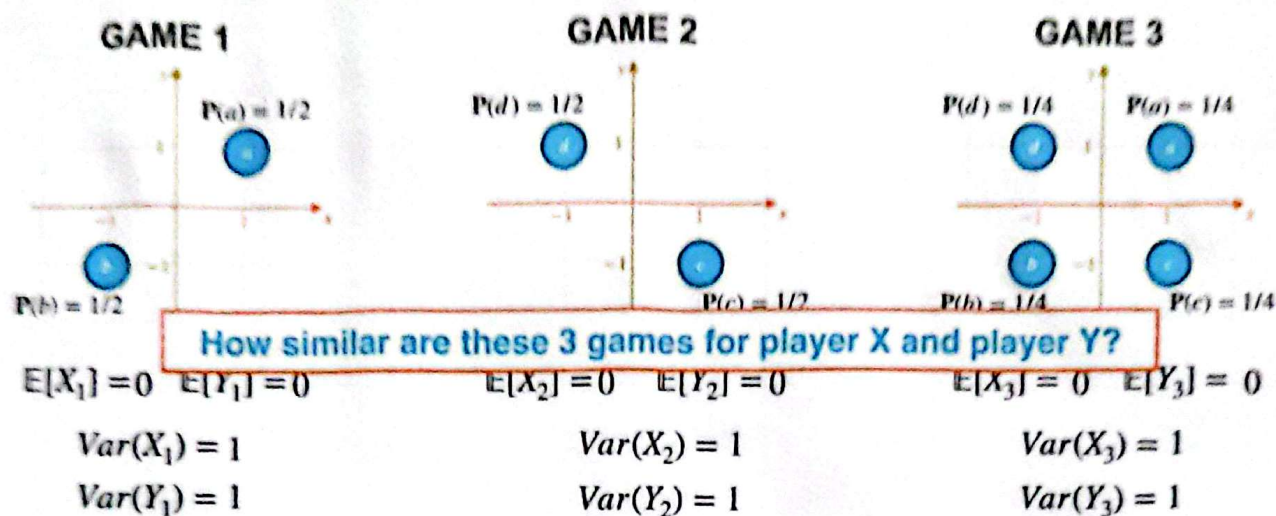


Age vs Naps per Day

Covariance < 0

$$\text{Cov}(x, y) = -7.45$$

Covariance of a Probability Distribution: Motivation



18) Covariance matrix

*] For 2 variables

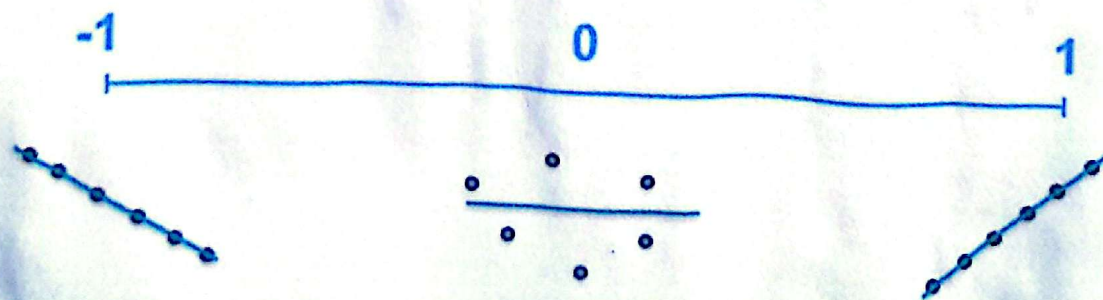
$$Cm = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix}_{2 \times 2}$$

*] For 3 variables

$$Cm = \begin{matrix} & X & Y & Z \\ \begin{matrix} X \\ Y \\ Z \end{matrix} & \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) & \text{Cov}(X, Z) \\ \text{Cov}(Y, X) & \text{Var}(Y) & \text{Cov}(Y, Z) \\ \text{Cov}(Z, X) & \text{Cov}(Z, Y) & \text{Var}(Z) \end{bmatrix} \end{matrix}$$

19) Correlation Coefficient

$$\text{Correlation Coefficient} = \frac{\text{Cov}(X, Y)}{\sigma_x \cdot \sigma_y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

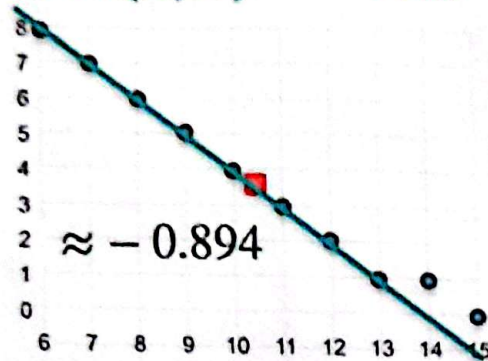


Age vs Naps per Day

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 7.57$$

$$\text{Cov}(X, Y) = -7.45$$



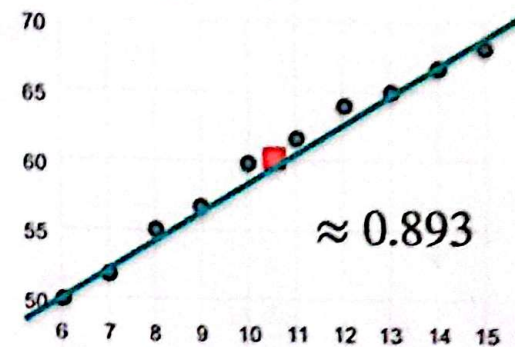
Age vs Naps per Day

Age vs Height

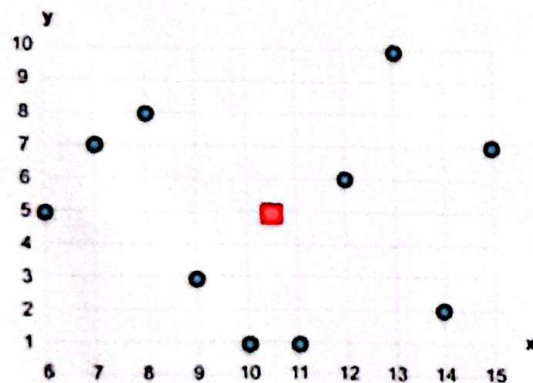
$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 39.56$$

$$\text{Cov}(X, Y) = 17$$



Age vs Height



Age vs Grades

$$\text{Var}(X) = 9.17$$

$$\text{Var}(Y) = 9.78$$

$$\text{Cov}(X, Y) = 0.1$$

Correlation
Coefficient

$$= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

$$= \frac{0.1}{\sqrt{9.17} \cdot \sqrt{9.78}}$$

$$\approx 0.01$$

20) Multivariate Gaussian Distribution

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-1/2 \frac{(x-\mu)^2}{\sigma^2}}$$

$$x = [x_1 \quad x_2 \quad \dots \quad x_n]^T$$

$$f_X(x_1, x_2, x_3, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)}$$

Covariance Matrix
spread of the bell

Rescaling /
standardization

$|\Sigma|$ determinant of the covariance matrix

* For univariate, we work with scalar values and variances

* For multivariate, we work with vectors and the covariance matrix